

Adora PINN — Step 6: Losses & derivative checks

As-built notes · Ivan Milic · 2026-09-20 · step 6 of pinn/plan.md

Step 6 assembles the three inversion losses against the step-5 testset and verifies their gradients w.r.t. the field params. Files [pinn/losses.py](#), [pinn/step6_losses.ipynb](#).

The losses (all differentiable through field $\rightarrow$ EOS $\rightarrow$ linear RT).

- **fit_loss** — χ^2 of synthesized Stokes vs observations on random pixels; uses `build_synth_cube_eos(..., forward_only=False)` so the gradient flows params $\rightarrow$ field $\rightarrow$ EOS $\rightarrow$ RT $\rightarrow$ Stokes.
- **he_loss** — hydrostatic equilibrium $dP/dz = -\rho g$ in **physical** units at random collocation points: dP/dz by autodiff of the field w.r.t. z (with the $z_{\text{norm}} \rightarrow z_{\text{phys}}$ chain rule), $\rho = n_{\text{H,tot}} \bar{m}$ amu from the EOS; relative residual.
- **boundary_loss** — soft $\log_{10}$ constraint on the top gas pressure ($z_{\text{norm}} = +1$).

Derivative check. `directional_fd` compares a central finite difference along a random direction to $\langle \nabla, v \rangle$: `rel_err` = fit 5×10^{-3} , HE 2×10^{-3} , boundary 4×10^{-4} , and every gradient is finite. A 150-step Adam descent on the total loss reduces it $9.6 \times 10^3 \rightarrow 1.8 \times 10^2$ ($\sim 50\times$) with Stokes V emerging — the gradients are usable.

Key fix. The check caught a **NaN fit-gradient at $B = 0$** — the inversion's starting point — from the singular Cartesian $\rightarrow (|B|, \gamma, \chi)$ map. `atmosphere_field.b_to_rt_angles` is now gradient-safe: soft-norm $\sqrt{\cdot + \epsilon^2}$ ($\epsilon=10^{-8}$), $\gamma = \text{atan2}(b_h^{\text{safe}}, B_z)$, and a double-where azimuth. $|B| \cos \gamma = B_z$ is still exact, so $\partial V / \partial B_z$ stays correct at $B=0$ (ϵ is physically negligible; steps 4–5 spectra unchanged).

Note. Here the fit pixels and collocation points are *fixed* (a clean function to finite-difference); the real inversion (step 7) resamples them every epoch.